Heterogeneity in RTA Trade Effects

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Abstract

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1 Introduction

A government weighing a trade agreement asks a question about itself: what will this agreement do to our trade? The empirical literature answers a different question: what have such agreements done, on average, across all pairs of countries that ever signed one. The two questions have the same answer only if agreements affect all pairs alike. If they do not, the average describes the treaty stock rather than any member of it, and the averaging itself does damage beyond answering the wrong question: it pools entities that appear related but differ qualitatively — in what they trade, in how their economies interlock, in what the agreement can reach — and smooths over the distinct causal stories that generated the data (Heckman, 2001). What the questioner wanted was precisely the contingency the pooling removes. Whether this substitution of one question for another is harmless is an empirical matter. This paper argues that in the flagship empirical model of trade it is not, and that its cost has been systematically underestimated because the tools that should reveal it are structurally unable to.

There are good reasons to expect the substitution to matter. An agreement changes the terms under which particular exporters in one economy meet particular buyers in another, and what follows depends on the complementarity of the two economies, the capacity behind their existing trade, and the quality of the match the agreement deepens — determinants no dataset records and no regressor represents. When relevant determinants go unobserved, the influence of the observed ones does not survive untouched: the response to a measured policy is mediated by, and contingent on, the unmeasured conditions under which it operates. Heterogeneity in treatment effects is the statistical name for this contingency — the form that dependence on the unmeasured takes when viewed from inside a regression, represented as variation in the effect precisely because it cannot be represented as a covariate (Pesaran and Smith, 1995). Homogeneity, by contrast, is the assertion that none of these conditions matter: that the same treaty raises trade between neighboring industrial economies and between distant, barely connected ones by an identical proportion. Framed this way, the common-effect specification is not a neutral default awaiting evidence against it. It is the strong assumption, and the burden it has quietly carried in applied work deserves examination rather than presumption.

The substitution would still be tolerable if the estimated coefficient were consumed as a number — reported, compared, filed. It is not. It is consumed as an input, and the chain it enters is nonlinear at every link. The agreement coefficient enters trade-cost functions exponentially, and through them the counterfactual general-equilibrium systems in which the gravity literature conducts its policy analysis, where prices, incomes, and multilateral resistances adjust jointly. Nonlinear transformations do not commute with averaging (Jensen, 1906): the outcome implied by the average effect is not the average of the outcomes implied by the effects, and under convexity, dispersion is not noise surrounding the answer but part of the answer, shifting levels rather than merely widening bands. Nor does the chain end at the model's output. The users of these counterfactuals do not care about a trade elasticity, or even a trade response, for their own sake; they care about what the response does to the things the analysis is meant to inform — revenues, adjustment burdens, the position of an economy vis-à-vis its partners — and that second mapping, from model output to consequence, is nonlinear once more. Two nested nonlinearities amplify whatever the point estimate conceals. Small differences in the estimate, and modest amounts of hidden dispersion around it, compound into large differences in the quantities the exercise exists to compute. Put plainly, a point estimate fed into a nonlinear model is an implicit distributional assumption — it asserts that the distribution of effects is degenerate at a point — and if the true distribution is dispersed, skewed, or tilted across the very pairs the counterfactual concerns, the model returns confident answers about a world that does not exist, with the confidence inherited from an assumption nobody stated.

Nor does the difficulty begin only where the estimate is used; it begins where the estimate is made. "The average effect" reads as a single object, but estimators do not compute unweighted means over pairs. Every fixed-effects structure weights the pairs by the variation left after its absorptions, and different structures — absorbing time, absorbing pair, absorbing country-by-time — leave different variation and therefore weight different pairs. Under homogeneity this is immaterial: all weightings of a constant agree, and the well-documented sensitivity of gravity estimates to specification can only be read as fragility, an embarrassment to be managed. Under heterogeneity the same sensitivity is a prediction. If effects vary systematically along dimensions the weights track, then correctly specified, individually defensible estimators must disagree, and the pattern of their disagreement is information about the shape of what is being averaged. Misread through the common-effect lens, that information becomes something worse than wasted: it becomes an invitation to choose among specifications by their conclusions, with each choice defensible and none interpretable.

The natural escape is to stop averaging: estimate the effect pair by pair and study the collection. But this route fails in a way that is more instructive than the failure of the average, because it fails while passing every test. The instruments used to police unit-level estimates — replication across subsamples, stability across time, the reliability of cross-sectional rankings — all measure persistence. They ask whether an estimate returns when we look again. Yet the dominant distortions in pair-level trade data are themselves persistent characteristics of pairs: biases rooted in volatility and drift do not wash out on a second look; they reappear, and a diagnostic that certifies whatever persists will certify them with the same authority it would certify the truth. Reliable and wrong is a possible state of an estimate, and it is a state the standard toolkit cannot detect from the inside. Confidence, if it is to be had, must be earned differently: not by observing that estimates replicate, but by identifying the mechanisms that would make artifacts replicate, and removing them.

Assembled, the exposures compound. Whoever relies on an average agreement effect is exposed three times over: the average answers a question about the treaty stock rather than the question asked; the nonlinear chain it feeds amplifies whatever it conceals; and the diagnostics that should sound the alarm are structurally silent. None of this is visible in the point estimate or its standard error, which may be small, stable, and beside the point. What would resolve all three exposures is the same object: the distribution of agreement effects across pairs — its center, its spread, its skew, and its variation across the kinds of pairs that policy distinguishes. The resolution is more than formal. Decisions made under uncertainty are rarely symmetric in their stakes: the quantities a decision-maker actually needs are the floor, the probability of being made worse off, and the realistic upside — and these are properties of a distribution's shape that no mean, however precisely estimated, contains. With the distribution in hand, the question a government asks has an answer in the form it was posed; the nonlinear model can be fed what it actually requires; and the disagreement among specifications becomes decomposable rather than embarrassing.

There is, moreover, a reason to expect this distribution to have structure rather than being an amorphous cloud of noise. Economic magnitudes evolve by compounding: growth arrives as relative change applied to an existing state, so that trade between two economies at any moment is the residue of multiplied increments rather than summed ones (Gibrat, 1931; Sutton, 1997; Jones, 2016; Beare and Toda, 2022). A policy acting on such a process plausibly acts the same way — scaling what is there rather than adding to it — and multiplicative action by unmeasured conditions carries testable signatures: lognormal outcomes, asymmetry in effects, effects graded in the size of what they act upon, and selection of agreements on the very conditions that shape them (Mitzenmacher, 2004). These signatures are the bridge between the motivation above and the evidence below: they are what the data would display if the contingency argued for in this introduction is real, and what they should not jointly display if it is not.

This paper estimates the distribution of agreement effects for the flagship empirical model of trade, and makes four contributions along the way. First, we derive the failure of the diagnostics formally: we show why naive pair-level estimation produces estimates that are simultaneously badly biased and nearly perfectly reliable, and we state the conditions under which no persistence-based diagnostic can distinguish artifact from effect. Second, we construct the corrected distribution, removing the identified biases in sequence and the remaining estimation noise by deconvolution; where the data are genuinely silent — on the boundary between transitory effect variation and treatment-coincident volatility — we report a set rather than an assumption-dependent point, which is the same honesty applied at the second moment that motivates the paper at the first. Third, we resolve the disagreement among specifications constructively, showing that the estimates standard structures produce are differently weighted averages of the one distribution we recover, so that the specification spread is explained rather than adjudicated. Fourth, we propagate the distribution through general equilibrium and show that its shape — not merely its mean or its variance — governs the decision-relevant answers: the floor, the downside probability, and the upside; imposing a convenient shape in its place manufactures conclusions, including apparent risks of loss, that the data do not contain.

Section 2 develops the mechanism and states the four predictions it commits to. Section 3 describes the data and the construction of the estimating population. Section **??** presents the naive estimator and its diagnostics, derives the reliability results as formal propositions, and sets out the corrections. Section **??** reports the corrected distribution, its variation across pair types, the reconciliation of the specification spread, and the identification boundary. Section **??** propagates the distribution through general equilibrium. Section **??** concludes and reports which of the four predictions survived. Proofs are collected in Appendix B.

2 Theoretical Framework

2.1 The Economic Primitive

Structural gravity supplies the accounting within which bilateral trade is organized: an exporter's supply capacity, an importer's expenditure, and the trade costs between them measured relative to the costs each faces with every other partner, all disciplined by general-equilibrium consistency (Anderson and Van Wincoop, 2003; Head and Mayer, 2014). That apparatus fixes the size and distance terms and, in the fixed-effects form now standard, absorbs whatever varies by exporter-year and by importer-year. What it does not fix is the pair. Two dyads matched on exporter capacity, importer expenditure and distance routinely trade at volumes that differ by more than the residual variation the model treats as incidental, and the systematic part of that difference is the object of this section. The question is not whether such variation exists but what kind of object generates it, because the answer determines the shape of what a policy acting on such pairs can produce.

Trade between two economies requires that a series of conditions hold at once. What one economy produces must be what the other can use, at the stage of processing at which it is offered; contracts written under one legal order must be enforceable, in practice and not only on paper, under the other; the pair must sit on shipping, payment and correspondent-banking networks that already carry traffic between them; and the administrative apparatus of customs classification, standards recognition and documentation must be in place to a depth sufficient for goods to move without prohibitive delay. These conditions are specific to the pair, are individually unobserved, and no dataset records them. Decisively, they do not substitute for one another (Kremer, 1993). A shipment occurs because every link in the chain holds; an unusually complementary production structure does not compensate for an unenforceable contract, and a well-functioning payments corridor does not compensate for goods the importer cannot use. Conditions of this kind enter the outcome multiplicatively rather than additively, and the compounding is not a modeling convenience but a restatement of what it means for a chain to hold. Multiplicativity is itself a consequence of heterogeneity under complementarity: when the value of an activity depends on which others are present, the combinations available expand as a function of those already realized, and growth that builds on the existing stock in this way is multiplicative by construction.

This structure is neither peculiar to trade nor unfamiliar. It is the production structure of the weak-link literature, in which output under complementarity is governed by the least functional element rather than by the average of the elements, so that a defect at any point in a chain reduces the value of competence everywhere else (Kremer, 1993; Jones, 2011). The same logic organizes the older capital-theoretic treatment of complementarity, in which the productivity of an addition to the capital stock depends on the composition of the stock it joins rather than on its own quantity (von Böhm-Bawerk, 1921; von Hayek, 1941; Lachmann, 1956). More broadly, economic processes are characterized by mechanisms that reinforce what is already in place: network effects and increasing returns that compound over time (Pietronero et al., 2001; Gabaix, 2009); agglomeration (Krugman, 1991); and learning-by-doing, in which capability is a function of accumulated activity (Arrow, 1962; Romer, 1986). In each case the marginal contribution of a factor is conditioned on the configuration of the others rather than being fixed. Taking the bilateral pair as the unit at which such configurations are realized, the residual variation left by structural gravity is what a multiplicative interaction of unobserved pair conditions would look like when viewed through a specification that has no place to put them.

2.2 From Primitive to Distribution

If bilateral outcomes are the product of many pair-specific factors that vary independently enough not to move as one, the logarithm of the outcome is a sum of many varying components, and the distribution of outcomes across pairs belongs to the lognormal family — Gibrat's law of proportionate effect applied to the dyad rather than the firm (Gibrat, 1931; Sutton, 1997). This is the mechanism's first prediction: (i) bilateral trade flows are distributed lognormally across pairs, with the right-tail weight that family implies. The prediction is genuine, in that a mechanism generating additively separable conditions would not produce it, but its discriminating power is limited and it would be dishonest to present it otherwise. Lognormal outcomes are overdetermined: random proportionate growth, the aggregation of heterogeneous sub-units, and several other generative stories produce the same signature, and distinguishing among them from the shape of a single cross-section is generally not possible (Mitzenmacher, 2004). Confirming (i) therefore removes rival mechanisms that predict something else while leaving standing every rival that predicts the same thing, which is why this paper does not rest its case on this signature, and why the mechanism must be made to commit to predictions about the treatment effect and not only about the outcome.

2.3 Non-Separable Treatment Effects

The empirical models used to estimate trade flows are themselves multiplicative. The workhorse specification writes the flow as $y_{ij} = \exp(\mathbf{x}_{ij}'\boldsymbol{\beta} + \epsilon_{ij})$, estimated by Poisson pseudo-maximum likelihood under $\mathbb{E}[e^{\epsilon_{ij}} \mid \mathbf{x}_{ij}] = 1$ (Santos Silva and Tenreyro, 2006). The functional form is multiplicative but the shock is not: ϵ_{ij} enters in parallel to the covariates, so that whatever the pair's unobserved conditions do to its trade, they do independently of the policy variable sitting in $\mathbf{x}_{ij}$. An agreement then shifts every pair's trade by the same proportion, and the pair conditions of Section 2.1 are confined to setting the level from which that common proportion is taken. This is a substantive restriction, not a normalization. It asserts that the conditions determining whether the chain holds have no bearing on what happens when one link in the chain is loosened.

The alternative is to let the covariates interact with the shock, $y_{ij} = \exp(\mathbf{x}_{ij}'(\boldsymbol{\beta}+\boldsymbol{\epsilon}_{ij}))$, under the corresponding normalization $\mathbb{E}[e^{\mathbf{x}_{ij}'\boldsymbol{\epsilon}_{ij}} \mid \mathbf{x}_{ij}] = 1$, which preserves the familiar conditional mean $\mathbb{E}[y_{ij} \mid \mathbf{x}_{ij}] = e^{\mathbf{x}_{ij}'\boldsymbol{\beta}}$ and which we maintain throughout. This form has two readings. Structurally, it is a random-coefficients model in which the effect of each covariate, the agreement among them, varies across bilateral pairs, with the variation governed by the distribution of $\boldsymbol{\epsilon}_{ij}$. Empirically, it generates heteroskedasticity of the kind long documented in trade data (Santos Silva and Tenreyro, 2006): pairs facing higher systematic trade costs are more exposed to idiosyncratic frictions in implementation, and therefore deviate more widely, in absolute terms, from what the model predicts for them. Written in terms of the object of interest, the agreement effect for a pair is

$$\theta_{ij} \;=\; f(A_{ij},\, c_{ij})\,, \tag{1}$$

where A_{ij} denotes the agreement and c_{ij} the pair's unobserved conditions — rather than a scalar θ common to all pairs. Neither argument of f is homogeneous. Agreements enter the empirical literature as a binary indicator, but the legal instruments so indicated range from tariff schedules with little else attached to architectures governing services, procurement, standards and dispute settlement, and treating these as one object imposes a second homogeneity assumption alongside the first; we do not attempt to separate the two sources of dispersion here, and note only that both are suppressed by the same specification and neither is consistent with a scalar θ. The economics of the modulation is the economics of a chain. An agreement does not create trade directly; it relaxes a constraint, lowering a tariff, recognizing a standard, binding a dispute procedure. Whether relaxing that constraint moves anything depends on whether the other links were holding, which is precisely what c_{ij} records. The effect is therefore modulated by the pair's conditions rather than added to them, and three further predictions follow from that modulation.

The first concerns shape. If the effect is the product of the relaxation and the conditions it acts through, the effect distribution inherits the compounding of the conditions and is therefore right-skewed rather than symmetric: (ii) most pairs gain modestly, a minority whose conditions are aligned gain multiplicatively, and the left tail is thin. The asymmetry has a specific source, and it is worth stating plainly because it is what distinguishes this prediction from a generic appeal to skewness. The upside is unbounded, in the sense that the conditions can reinforce one another without limit; the downside is bounded, because an agreement that fails to bind is inert rather than destructive. A pair whose chain was broken before the agreement, and remains broken after it, records no effect — it does not record a loss. There are channels through which an agreement could reduce a particular pair's trade, diversion toward a third partner most obviously, but these operate against the floor rather than mirroring the upside, so the mechanism predicts an effect distribution with a short left tail and a long right one.

The second concerns gradation, and it is the prediction that gives the mechanism purchase on observable data. If an agreement works by relaxing constraints, its effect is largest where the constraints bind hardest. But the constraints a treaty addresses are discrete rather than continuous: a tariff line, a standards recognition, a documentation requirement. Each either still obstructs the pair or has already been surmounted, and a pair trading at volume through established channels has, by revealed behavior, already surmounted them — the fixed costs of compliance are sunk, the customs relationships exist, and the trade would not be occurring otherwise. Relaxing what has already been worked around changes little. The prediction is therefore that (iii) effects are graded in pair size and prior integration, but not smoothly: they are large where the treaty's constraints still bind and fall to a common floor once they do not. What the mechanism forbids is continued decline among pairs that have already exhausted the margin, since beyond exhaustion there is nothing left to grade. This is sharper than (i) and (ii), because it names both a direction and a shape against variables the data contain, and it is falsified not only by a positive slope but by a smoothly monotone negative one.

The third concerns who signs. The conditions in c_{ij} are not observed by the econometrician, but they are observed, imperfectly and in the aggregate, by the governments and the exporters who lobby them. Agreements are negotiated where there is something to negotiate over, which is to say where enough of the chain already holds that the parties can foresee gains from relaxing what remains. The prediction is that (iv) agreements are selected on the same unobserved conditions that modulate their effects: treated pairs are drawn from the stable end of the condition distribution rather than at random from it. This is assortative matching of the kind that complementarity produces generally, where partners of similar reliability are matched together in equilibrium precisely because reliability is complementary (Kremer, 1993), applied to governments choosing whom to negotiate with. Two consequences follow, and they should be stated together because they pull in different directions. Any estimate recovered from treated pairs is an effect on self-selected matches and does not license extrapolation to a pair that has not signed, which limits the external reach of what the sections below can report. But the same selection is a testable implication rather than only a limitation: it predicts that the treated population is more stable in its pre-agreement behavior than the untreated, and that prediction can be checked without observing c_{ij}.

Predictions (iii) and (iv) may appear to pull against one another, the first locating the largest effects among small and thinly integrated pairs, the second locating agreements among pairs whose conditions already hold. They operate, however, on disjoint parts of c_{ij}. Selection runs on the links a treaty does not address: contractual enforceability, payment and correspondent-banking infrastructure, the administrative capacity of the counterparty — the conditions that make gains foreseeable and therefore worth the cost of negotiating. Gradation runs on the links a treaty does address: tariffs, administrative barriers at the border, standards recognition — the constraints still binding that an agreement can relax. A pair may have a functioning non-policy chain, and so be a plausible negotiating partner, while its policy-addressable constraints remain fully binding and its realized volume correspondingly low. That combination describes a small, thinly integrated pair with a competent counterparty, and it is not unusual. The two predictions are therefore compatible, and their conjunction is sharper than either alone: the mechanism predicts that the treated population is drawn from the stable end of the condition distribution and not from the large end, and stability and size are separable in the data.

The distinction between the non-separable form and the additive alternative, in which the shock enters in parallel to the covariates rather than multiplying them, is consequential rather than conventional: only the non-separable form generates dispersion through the interaction of the shock with treatment, and only it implies that the dispersion of the outcome varies with treatment. That implication is testable directly, on pairs observed before and after adoption, and the test bears the weight of everything that follows: if the additive form survives it, the modulation in Equation (1) is a redescription rather than a mechanism, and the corrections built upon it are unmotivated. We defer the argument for the non-separable structure, and the test that distinguishes it from the additive alternative, to Section **??** [X-REF: dispersion subsection], where it underpins the analysis of how the adjustment interacts with the fixed-effects structure.

2.4 The Evidentiary Standard

Each of these four predictions is weak taken alone, and we ask the reader to hold the argument to the standard the mechanism actually meets rather than to one it does not. Lognormal outcomes are overdetermined, as Section 2.2 conceded. Right-skewed effects are consistent with any story in which gains are bounded below and unbounded above. A size gradient is consistent with diminishing returns of a wholly different origin, or with measurement error correlated with volume. Selection on unobservables is consistent with almost any account of why governments negotiate. What the mechanism claims is not that any one of these is decisive but that their conjunction is jointly predicted: lognormal outcomes *and* right-skewed effects *and* effects graded in size and prior integration *and* selection on the conditions that do the modulating. Each rival that explains one of these comfortably explains at least one of the others badly or not at all, and to our knowledge no single alternative currently on the table predicts all four together. This is abduction, not proof: the conjunction is evidence for the mechanism in the sense that the mechanism would render it unsurprising and its rivals would not, and a mechanism we have not considered may do as well or better. Sections **??** and **??** test each prediction in turn — (i) in the distributional diagnostics, (ii) and (iii) in the corrected effect distribution and its variation across pair types, and (iv) in the composition of the treated population — and Section **??** reports which survived.

3 Data and the Estimating Population

Section 2.4 committed the mechanism to four predictions and assigned each to a later section. Those assignments are obligations on the data as much as on the method. A prediction about the shape of outcomes, about gradation in size and prior integration, and about the composition of the treated population can be tested only against a population whose construction is fixed in advance and stated in full, since a construction chosen after the estimates are seen can accommodate any of them. This section fixes it. We describe the trade panel and the two configurations in which it is used, the agreement indicator and how adoption is dated from it, the rules that select the pairs on which pair-level effects are computed, and the populations from which the counterfactual is built. Nothing here reports an estimate.

3.1 Trade Flows

Bilateral trade comes from the International Trade and Production Database for Estimation, Release 2 (Borchert et al., 2022), aggregated to total goods trade for each ordered exporter–importer pair and year, in millions of current United States dollars. The panel runs from 1988 to 2019 , which is the window over which the release's goods sectors are jointly available; its services industries begin only in 2000 and are outside the panel by coverage rather than by choice. After the restrictions described next, the estimation panel contains 794,720 observations across 46,803 ordered pairs , of which the terminal cross-section contributes 31,650 . These counts and the year bounds are frozen: every script verifies them against the source file before computing anything and halts if they do not reproduce.

Two configurations of this panel are used, and we state both here. Estimation of the agreement effect uses positive international flows: observations in which exporter and importer coincide are excluded, and observations recording zero trade are excluded with them. The first follows from what is being estimated — an agreement acts on trade between two countries, and a country's trade with itself carries no agreement — and the second from the pair-level construction, which takes logarithms of realized flows and their counterfactual counterparts. The general-equilibrium analysis of Section **??** re-introduces domestic flows, because its accounting requires them: an expenditure system in which each economy's outlay must exhaust across all sources cannot be closed on international flows alone. The two are not competing samples but the input requirements of two different calculations.

One property of the release deserves recording where the exclusions are stated rather than buried. Missing international flows in Release 2 are set to zero on the reasoning that reported trade is comprehensive and the bilateral matrix is genuinely sparse. A zero here is therefore an observed absence of recorded trade rather than a measured value of nothing, and dropping zeros drops both. We take no position on the imputation and note only that it makes the boundary between a pair that does not trade and a pair whose trade is unrecorded less sharp than the data's appearance suggests.

The outcome is the gross value of goods shipped from one member of the pair to the other. This is the right object rather than merely the available one. The treaty is a bilateral instrument: it is signed by two governments, it alters the terms on which exporters in one meet buyers in the other, and the unit at which its provisions bind is the pair. An outcome on any other unit would either aggregate across pairs the treaty treats differently or decompose the pair into components the treaty does not address. Value-added measures reallocate part of a bilateral relationship's response to the third countries supplying its inputs, which is informative about where production sits but not about what the agreement did to the relationship it governs; netting flows within a pair discards the directional asymmetry the provisions themselves distinguish. Gross directed flows keep estimand and instrument on one unit.

3.2 Agreements

Agreement status is the binary indicator of Egger and Larch (Egger and Larch, 2008), recording for each pair-year whether a reciprocal agreement is in force. Pairs are classified from the time path of this indicator alone. A pair is a *switcher* if the indicator makes exactly one transition from zero to one and none from one to zero; its adoption year is the first year the indicator equals one. Pairs whose indicator equals one in every year are always-treated: they carry no within-pair variation and cannot contribute to a within-pair design, whatever their weight in world trade. Pairs recording more than one transition are excluded, because adoption is not well defined for them and a pair-level effect computed across their regimes would average over the very heterogeneity the paper measures. Together these exclusions are large: 6,339 of 46,803 pairs are single switchers , and the 40,464 removed are overwhelmingly never-treated.

The indicator is binary, and Section 2.3 has already observed that a single indicator bundles legal instruments ranging from bare tariff schedules to architectures governing services, procurement, standards and dispute settlement; the construction described here does not separate variation originating in that bundling from variation originating in pair conditions, and we make no claim about their relative contribution.

3.3 The Estimating Population

The design is within-pair: each pair's effect is identified from the comparison of its own realized trade after adoption with what the fitted model implies for it in the absence of the agreement, so the pair is its own control and no cross-pair comparison enters the pair-level estimate. Every selection rule follows from that design, and each is stated with the reason it exists rather than as a threshold applied by convention. Table 1 reports the pairs surviving each rule in the order applied.

The rules are: (a) *switchers only*, since the within-pair comparison requires observation on both sides of adoption, and always-treated and multi-switcher pairs are excluded by Section 3.2. (b) *adoption within 1991–2016*: a pair adopting in the panel's first or last years cannot satisfy the window requirements below, and the range makes the boundary explicit rather than implicit; it removes 753 pairs . (c) *a symmetric one-year exclusion band around adoption.* The pre-period ends at adoption − 2 and the post-period begins at adoption + 2; the adoption year and both flanking years are assigned to neither side. Agreements are announced before they enter force and implemented in stages after, so trade in the immediate neighbourhood of the switch belongs cleanly to neither regime. The band is symmetric by construction, and the same band governs the comparison populations of Section 3.4, so that treated and untreated pairs are never compared across differently drawn windows.

Two further rules concern estimability and window length. (d) *a usable counterfactual*:

a pair enters only if the imputation model of Section **??** yields a prediction for its post-adoption cells, which requires that its own fixed effect be identified from untreated data. This removes 576 pairs ; because identification binds at the pair fixed effect rather than cell by cell, every retained pair has a counterfactual for all of its post-adoption years, and no retained pair is estimated on a truncated window. (e) *at least three pre- and three post-adoption years of positive trade*, outside the exclusion band, removing 353 and 475 pairs respectively . A pair-level effect estimated on one or two years is dominated by whichever shocks fall in them, and the minimum buys enough within-pair observations for the effect and its sampling error to be separately estimable. Robustness across the threshold grid is reported in Appendix A [X-REF: threshold grid].

Rule (e) carries a consequence that should be stated rather than left to be discovered. A pair recording no positive trade in any pre-adoption year has no pre-period and is removed by the rule mechanically; there is no separate extensive-margin exclusion because none is needed. The effect this paper estimates is therefore an intensive-margin effect — what an agreement does to trade between countries already trading — and it is silent on the creation of trading relationships where none existed. That is a real restriction on the estimand's reach, since the excluded pairs are precisely those for which an agreement might matter most in proportional terms.

Applying these rules yields the canonical population of 4,182 pairs , with adoption years from 1992 to 2015. The population is economically substantial: its members account for 22.1 percent of world international goods trade in 2015 . Its composition, however, deserves comment. The largest members by pre-adoption trade are the Korea–United States agreement of 2012, the Australia–Japan and Australia–China agreements of 2015, the China–Hong Kong arrangement of 2003, and the Japanese agreements with Thailand, Indonesia and Singapore ; the ten largest directed pairs account for 34.3 percent of the population's pre-adoption trade. Conspicuously absent are the North American and European agreements a reader might expect. They are absent by construction rather than by selection: Canada and the United States were already in force under their 1989 agreement when the panel opens, and intra-European pairs are in force throughout, so all are always-treated and carry no within-pair variation to identify from. The population is thus weighted toward the Asia-Pacific agreements of the 2000s and 2010s, and results should be read as speaking to agreements of that kind.

With the population fixed, the object of estimation can be stated exactly. It is the distribution across adopting pairs of the pair-level multiplicative effect of a new agreement on bilateral trade, for pairs with pre- and post-adoption trading histories.

3.4 The Comparison Populations

Two further populations are constructed here and used later. The first is the never-treated pairs: those whose indicator is zero throughout, to which pseudo-adoption years are assigned under the same band and minimum-length rules that govern the switchers. Because nothing happens to these pairs at their assigned dates, whatever a pair-level effect returns for them is by construction not an agreement effect, and the population therefore serves as a reservoir against which the machinery can be calibrated. The second is the switchers' own pre-adoption periods, which record how each pair behaves in the absence of the treatment it later receives, and so allow the pairs to calibrate themselves rather than each other.

These two populations have a further role that belongs here rather than later, because it determines which observations are used at all. Their union — every year of every never-treated pair, together with every pre-band year of every switcher — is the sample on which the counterfactual is fitted. No cell in which an agreement is in force enters that fit. The consequence is that a switcher's post-adoption counterfactual is a prediction for years the fitting sample never saw, rather than a fitted value for years it did; what that buys, and what it costs, is the business of Section **??**. We record here only the sample definition, since it is a property of the data construction and not of the estimator applied to it.

Table 1: Construction of the estimating population

	Rule	Pairs	Removed
	All ordered pairs, international, positive trade	46,803	—
(a)	Single switchers (one $0 \to 1$ transition)	6,339	40,464
(b)	Adoption year in $[1991, 2016]$	5,586	753
(d)	Usable counterfactual from untreated data	5,010	576
(e1)	≥ 3 pre years, outside the band	4,657	353
(e2)	≥ 3 post years, outside the band	4,182	475
	Canonical population	**4,182**	

Notes. Rules are applied in the order shown. Rule (c), the symmetric one-year exclusion band, is not a separate row: it defines the pre- and post-periods over which (e1) and (e2) count years, so its effect is embedded in those two rows. Pre years are those before adoption $-$ 1 and post years those after adoption + 1, both requiring positive trade. Panel totals are frozen and verified at load . Source: .

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A Additional Tables and Figures

B Proofs

C Propositions

Notation and assumptions

y_{ijt} observed trade; $\hat{y}_{ijt}$ the PPML-imputed counterfactual; $\eta_{ijt} := y_{ijt}/\hat{y}_{ijt}$; $g_{ijt} := \log \eta_{ijt}$. Definitions: $\theta^A_{ij} := T^{-1} \sum_{t \in \text{post}} g_{ijt}$; $\theta^B_{ij} := \log\big(\sum_t y_{ijt} / \sum_t \hat{y}_{ijt} \big)$ over the post window.

Assumption 1 (A1)**.** *$y_{ijt} = \hat{y}_{ijt}\eta_{ijt}$ with $\mathbb{E}[\eta_{ijt}] = 1$.*

Assumption 2 (A2)**.** *$\log \eta_{ijt} \sim N(-\sigma^2_{ij}/2, \sigma^2_{ij})$, i.i.d. over t within pair.*

Assumption 3 (A3)**.** *σ^2_{ij} varies across pairs and is a persistent pair trait.*

Proposition 1 (Jensen bias and its reliability)**.** *Under (A1)–(A2) with zero treatment effect:* (a) *$\mathbb{E}[\theta^A_{ij} \mid ij] = -\sigma^2_{ij}/2$ exactly.* (b) *Write $R_{ij} = \sum_t w_t \eta_{ijt}$ with $w_t = \hat{y}_{ijt} / \sum_s \hat{y}_{ijs}$. Then $\mathbb{E}[\theta^B_{ij}] = -\frac{1}{2}\operatorname{Var}(R_{ij}) + r_{ij}$, where $\operatorname{Var}(R_{ij}) = \operatorname{Var}(\eta_{ij}) \sum_t w_t^2$ (equal weights: $\operatorname{Var}(\eta_{ij})/T$), $\operatorname{Var}(\eta_{ij}) = e^{\sigma^2_{ij}} - 1$, and the remainder r_{ij} collects terms of order $\mathbb{E}[(R-1)^3]$ and higher; the expansion requires $\operatorname{Var}(R_{ij}) \ll 1$.* (c) *For split-half estimates on disjoint post halves of length T_h,*

$$\operatorname{Corr}\big(\theta^{A,(1)}, \theta^{A,(2)}\big) = \frac{\operatorname{Var}_{pairs}(\sigma^2_{ij})/4}{\operatorname{Var}_{pairs}(\sigma^2_{ij})/4 + \mathbb{E}[\sigma^2_{ij}]/T_h},$$

exactly under (A2).

Proof. (a) Immediate from (A2). (b) Second-order expansion of $\log R$ about $R = 1$ with $\mathbb{E}[R] = 1$; the weight identity and $\operatorname{Var}(\eta) = e^{\sigma^2} - 1$ (from $\mathbb{E}[\eta^2] = e^{\sigma^2}$ under (A2)) give the display. (c) Conditional on the pair, $\theta^{A,(h)} = -\sigma^2_{ij}/2 + \bar{\varepsilon}^{(h)}$ with $\bar{\varepsilon}^{(h)} \sim (0, \sigma^2_{ij}/T_h)$ independent across halves; decompose variance and covariance across pairs. □

Corollary 1 (Reliability paradox)**.** *Let b_{ij} be* any *persistent per-pair bias with cross-pair variance* $\operatorname{Var}(b)$*; then under zero effects $r_{\text{split}} = \operatorname{Var}(b)/\big(\operatorname{Var}(b) + \mathbb{E}[\sigma^2]/T_h\big)$. High split-half reliability certifies persistence, not validity: any dispersed, persistent artifact (Jensen, drift) passes.*

Remark (Empirical scale; validity of the expansions)**.** *At the ledgered dispersion ($\mathbb{E}[\sigma^2] \approx 1.42$, i.e. mean placebo -0.71 under A),* $\operatorname{Var}(\eta) \approx 3.1$*: the generic Taylor form $-\operatorname{Var}(\eta)/2$ is* invalid *for θ^A (the exact lognormal value applies), and for θ^B at $T = 10$, $\operatorname{Var}(R) \approx 0.31$, so $-\operatorname{Var}(R)/2 = -0.156$ is a rough guide only. At the empirical scale ($\sigma = 1.19$, $T = 10$),* *simulation gives* $\mathbb{E}[\theta^B] = -0.117$ *against the second-order approximation* -0.156*: the expansion overstates the bias magnitude by roughly a third, the positive third-order (skewness) term partially offsetting the variance term. The approximation is thus a conservative guide at this scale. The ledgered B-placebo band* $[-0.23, -0.19]$ *additionally contains drift (Proposition 2). Consistency check [V1c]: the ledgered* $(r, \mathbb{E}[\sigma^2]) = (0.62, 1.42)$ *are jointly consistent with the part-(c) formula for* $\mathrm{Var}(\sigma^2) \approx 1.85$ *(CV of* $\sigma^2_{ij} \approx 1$*) — a plausible but not independently measured dispersion; [V1c] scans* $\mathrm{Var}(\sigma^2) \in [1, 3]$ *at the ledgered* $\mathbb{E}[\sigma^2] = 1.42$*: the implied r ranges* 0.47–0.73*, with the ledgered* $r = 0.62$ *obtained at* $\mathrm{Var}(\sigma^2) \approx 1.85$ *(a coefficient of variation of* σ^2_{ij} *near one) — consistent with, though not an independent measurement of, the observed reliability.*

Proposition 2 (Within-pair drift). *Relax (A2) to* $\log \eta_{ijt} = -\sigma^2_{ij}/2 + \delta_{ij}\,(t - \bar{t}_{ij}) + u_{ijt}$, $u_{ijt} \overset{iid}{\sim} N(0, \sigma^2_{ij})$*, with* δ_{ij} *a persistent pair-specific drift and* $\bar{t}_{ij}$ *the pair's sample midpoint (the pair fixed effect centers the level, not the trend). For a never-treated pair with span* $T = T_{\text{pre}} + T_{\text{post}}$ *and post window equal to the last* T_{post} *years:* (a) $\overline{(t-\bar{t})}^{\,\text{post}} = T_{\text{pre}}/2$*, hence to first order* $\mathbb{E}[\theta^A_{ij}] = -\sigma^2_{ij}/2 + \delta_{ij} T_{\text{pre}}/2$ *(and* θ^B *inherits the same drift term with* $\hat{y}$*-weighted window mean); the expression depends on window geometry only, never on calendar time.* (b) *The drift term is a persistent pair trait and passes split-half reliability by Corollary 1.* (c) *Now let true effects* $\theta_{ij} \sim (\mu, \tau^2)$ *enter the post window. If* δ_{ij} *correlates with pair size s, the Definition-D correction (subtracting the size-cell placebo mean) removes* $\mathbb{E}[\delta_{ij} \mid s]\, T_{\text{pre}}/2$ *but not the within-cell dispersion:*

$$\mathrm{Var}(\theta^D \mid s) \;=\; \tau^2 + \big(T_{\text{pre}}/2\big)^2\, \mathrm{Var}(\delta_{ij} \mid s) + \frac{\mathbb{E}[\sigma^2_{ij} \mid s]}{T_{\text{post}}},$$

so any deconvolution recovers τ^2 *jointly with the within-cell drift variance unless the null design carries matching window geometry.*

Proof. (a) The post-window mean of t is $T_{\text{pre}} + (T_{\text{post}}+1)/2$; subtracting $\bar{t} = (T+1)/2$ gives $T_{\text{pre}}/2$. Calendar year enters nowhere. (b) Apply Corollary 1 with $b_{ij} = -\sigma^2_{ij}/2 + \delta_{ij} T_{\text{pre}}/2$. (c) Decompose $\theta^D = \theta_{ij} + \big(\delta_{ij} - \mathbb{E}[\delta \mid s]\big) T_{\text{pre}}/2 + \bar{u}$, take variances conditional on s. □

Remark. *Part (a) explains the ledgered calendar invariance of the placebo bias (*$R^2 <$ 1% *on pseudo-year dummies) and, with size-correlated* δ*, the size-graded placebo means (*-1.07 *smallest decile); [V2a]–[V2b] verify the formula and the invariance in simulation. The ledgered M1 adjudication — stationary resampling produced* $b \in [0.03, 0.12]$*, incapable of generating the placebo pattern — is the empirical counterpart of the exclusion of (A2)-only mechanisms.*

Proposition 3 (Identification and its boundary). *Let pre-window gaps be* $g_{ijt} = \sigma_{ij} u_{ijt}$ *(mean-adjusted) and post-window gaps* $g_{ijt} = \theta_{ij} + \nu_{ijt} + \sigma_{ij}\rho\, u_{ijt}$*, with* $u_{ijt} \overset{iid}{\sim} N(0, 1)$*, transitory effect variation* $\nu_{ijt} \overset{iid}{\sim} N(0, \omega^2)$ *independent of u, persistent effects* $\theta_{ij} \sim (\mu, \tau^2)$*,* *and post/pre volatility ratio $\rho \geq 0$. Define the per-pair observables: pre variance V_{ij}^{pre}, centered post variance V_{ij}^{post}, and $\hat{\theta}_{ij} = \bar{g}^{\text{post}}$. Then:* (a) (Identification of τ^2.) $\mathbb{E}[V_{ij}^{\text{post}}] = \sigma_{ij}^2\rho^2 + \omega^2$, $\text{Var}(\hat{\theta}_{ij} \mid ij) = V_{ij}^{\text{post}}/T_{\text{post}}$ *exactly, hence*

$$\tau^2 = \text{Var}_{pairs}(\hat{\theta}) - \mathbb{E}\big[V^{\text{post}}\big]/T_{\text{post}},$$

with all right-hand quantities estimable (the within-pair sample variance is unbiased for V^{post}). (b) (Exact non-identification of (ω^2, ρ).) *Under the stated normality, the post gaps are, conditionally on the pair, i.i.d. $N(\theta_{ij},\, \sigma_{ij}^2\rho^2 + \omega^2)$: the parameters enter the likelihood only through the composite $\sigma^2\rho^2 + \omega^2$. Any two parameterizations with equal composites generate* identical *data distributions; the split between transitory effect heterogeneity and treatment-coincident volatility change is a modeling label, not an estimable quantity. Absent normality, separation requires within-pair moments of order ≥ 3, whose sampling noise at $T_{\text{post}} \approx 10$ (*SD *of sample excess kurtosis* $\approx \sqrt{24/T} \approx 1.5$) *dominates plausible population differences.*

Corollary 2 (Operational variance ratios over-subtract)**.** *Define $\hat{\rho}_{\text{op}}^2 := V^{\text{post}}/V^{\text{pre}} = \rho^2 + \omega^2/\sigma^2$. A "post-calibrated" null that scales a* pre-based *noise variance of window length $T_0 < T_{\text{post}}$ by $\hat{\rho}_{\text{op}}^2$ subtracts $\kappa_w V^{\text{post}}/T_{\text{post}}$ with window factor $\kappa_w := T_{\text{post}}/T_0 > 1$, biasing the recovered variance by $-(\kappa_w - 1)\,\mathbb{E}[V^{\text{post}}]/T_{\text{post}}$. With the ledgered $\kappa_w \approx 2.4$, a single σ calibrated from the ledgered injection scenario (ii) (*$0.40 \to 0.35$*) predicts the scenario-(iv) collapse (*$0.40 \to\approx 0$, *reported* 0.06*): [V3c] confirms: calibrating a single $\sigma \approx 0.52$ to the scenario-(ii) recovery (*$0.40 \to 0.35$*) predicts the scenario-(iv) collapse to the floor (*$0.40 \to 0$*; ledgered* 0.06*), by formula and by direct simulation of the mis-windowed estimator..*

Remark (The empirical bracket, organized)**.** *Part (a) is precisely the window-matched estimator used in the ledgered V1 computation. The ledgered bracket* $[0.39, 0.46]$ *corresponds to the two admissible noise attributions: subtracting $\mathbb{E}[\sigma^2]/T$ (pre-calibrated, conservative when the measured $\hat{\rho}_{\text{op}}^2 < 1$) yields* 0.39*; subtracting $\mathbb{E}[V^{\text{post}}]/T$ per part (a) yields* 0.46*. Under the stated model the latter is the identified value of the* persistent *heterogeneity τ; the transitory slice ω^2/T remains, by part (b), inseparable from volatility change at any sample size — the boundary of what these data permit anyone to learn.*